# Fractions and Percentages

Number · Edexcel IGCSE Mathematics A (4MA1) · Year 10 · worksheet

A+ Maths Hub

| Name | Class | Date |
|---|---|---|

**Answer ALL questions.**

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 154 minutes · Total marks: 154

## KEY RULES

| | |
|---|---|
| **Add / subtract** | Common denominator, then add the numerators |
| **Mixed numbers** | Change to improper fractions first |
| **Multiply** | Multiply tops and bottoms: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$ |
| **Divide** | Flip the second fraction and multiply |
| **Percentage change** | Multiplier: increase 15% → × 1.15 ; decrease 15% → × 0.85 |
| **Reverse percentage** | Divide by the multiplier: original $= \frac{\text{new}}{\text{multiplier}}$ |
| **Compound interest** | $A = P\left(1 + \frac{r}{100}\right)^n$ |

## WORKED EXAMPLE 1

[3 marks]

**Work out $2\frac{1}{3} \div 1\frac{3}{4}$. Give your answer as a mixed number.**

| | | |
|---|---|---|
| **Step 1** | Change to improper fractions | $\frac{7}{3} \div \frac{7}{4}$ |
| **Step 2** | Flip the second fraction, multiply | $\frac{7}{3} \times \frac{4}{7}$ |
| **Step 3** | Cancel the 7s | $\frac{4}{3}$ |
| **Step 4** | Write as a mixed number | $1\frac{1}{3}$ |

**Answer:** $1\frac{1}{3}$

## WORKED EXAMPLE 2

[3 marks]

**In a sale all prices are reduced by 20%. The sale price of a coat is $64. Work out the original price.**

| | | |
|---|---|---|
| **Step 1** | Sale price is 80% of the original | $80\% = 64$ |
| **Step 2** | Find 1% | $1\% = 64 \div 80 = 0.8$ |
| **Step 3** | Find 100% | $100\% = 0.8 \times 100 = 80$ |

**Answer:** $80

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## WORKED EXAMPLE 3

[3 marks]

**$2000 is invested at 3% compound interest per year for 4 years. Work out the total interest.**

| | | |
|---|---|---|
| **Step 1** | Multiplier for a 3% increase | 1.03 |
| **Step 2** | Value after 4 years | $2000 \times 1.03^4 = 2251.02$ |
| **Step 3** | Interest = final value minus start | $2251.02 - 2000$ |

**Answer:** $251.02

**1** Work out $\frac{2}{5} + \frac{1}{3}$

**(Total for question = 1 mark)**

*Practice question written for this worksheet (not from a past paper)*

**2** Work out $\frac{3}{4} - \frac{2}{7}$

**(Total for question = 1 mark)**

*Practice question written for this worksheet (not from a past paper)*

**3** Work out $2\frac{1}{2} + 1\frac{3}{4}$

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**4** Work out $3\frac{1}{3} - 1\frac{4}{5}$

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

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**5** Work out $\frac{2}{3} \times \frac{9}{10}$

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**6** Work out $1\frac{1}{2} \times 2\frac{2}{3}$

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**7** Work out $\frac{3}{5} \div \frac{9}{10}$

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**8** Work out $2\frac{1}{4} \div 1\frac{1}{2}$. Give your answer as a mixed number.

**(Total for question = 3 marks)**

*Practice question written for this worksheet (not from a past paper)*

**9** Pam spends $\frac{1}{3}$ of her money on rent and $\frac{1}{4}$ on food.

What fraction of her money is left?

**(Total for question = 3 marks)**

*Practice question written for this worksheet (not from a past paper)*

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**10** $\frac{3}{8}$ of a number is 24. Work out the number.

**(Total for question = 3 marks)**

*Practice question written for this worksheet (not from a past paper)*

**11** In a class, $\frac{2}{5}$ of the students are boys.
$\frac{3}{4}$ of the boys wear glasses.
What fraction of the class are boys who wear glasses?

**(Total for question = 3 marks)**

*Practice question written for this worksheet (not from a past paper)*

**12** **4** Sara sells water bottles.

In July, she sold 450 water bottles.
In August, she sold 612 water bottles.

Work out the percentage increase in the number of water bottles she sold from July to August.

.....%

**(Total for Question 4 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 4*

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13 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.
This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

**(Total for Question 4 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 4*

14 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.

Work out the normal price of the watch.

..... euros

**(Total for Question 5 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 5*

15 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

**(Total for Question 7 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 7*

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**16** **8** A golf club increases the cost of membership by 17%
After the increase, the cost of membership is 3276 zloty.

Work out the cost of membership before the increase.

..... zloty

**(Total for Question 8 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 8*

**17** **8** In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

..... rupees

**(Total for Question 8 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 8*

**18** **8** Jane bought a new car for $18 000
The car depreciates in value by 15% each year.

Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest $

$.....

**(Total for Question 8 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 8*

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**19** Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.

Work out the number of points that Hari scored playing the game on Monday

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 9*

**20** **9** In a sale, normal prices are reduced by 15%

The sale price of a dishwasher is 612 Swiss francs.

Work out the normal price of the dishwasher.

..... Swiss francs

**(Total for Question 9 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 9*

**21** **9** Nisha invests 20 000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.
Give your answer correct to the nearest euro.

..... euros

**(Total for Question 9 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 9*

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**22** 9 Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.
Give your answer correct to the nearest koruna.

..... koruna

**(Total for Question 9 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 9*

**23** Arwa invests 200 000 yen in a savings account for 3 years.
She gets 1.4% per year compound interest.

Work out how much money Arwa will have in her savings account at the end of 3 years.
Give your answer correct to the nearest yen.

..... yen

**(Total for question = 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 10*

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**24** 3 Joshua buys a car for $12 500

He sells the car to Nina.

Nina pays

- a deposit of $1500
- followed by 36 monthly payments of $450

Work out Joshua's percentage profit.

.....%

**(Total for Question 3 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 3*

**25** 3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.
Give your answer correct to 3 significant figures.

.....%

**(Total for Question 3 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 3*

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**26** **4** Divya and Yuan each pay for a holiday at a special offer price.

| **Divya's holiday** | **Yuan's holiday** |
|---|---|
| Normal price: \$1600 | Normal price: \$1400 |
| Special offer:<br>16% off the normal price | Special offer:<br>$k$% off the normal price |

The amount that Divya pays is the same as the amount that Yuan pays.

Work out the value of $k$

$k =$ .....

**(Total for Question 4 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 4*

**27** **6** Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

| ticket **A** | ticket **B** |
|---|---|
| $\frac{1}{6}$ off normal price | 20% off normal price |

The sale price of ticket **A** is 140 euros.
The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

**(Total for Question 6 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 6*

**28** Shop **A** and Shop **B** have offers for buying the same type of laptop.

The normal price of the laptop in Shop **A** is different to the normal price of the laptop in Shop **B**

| **Shop A** | **Shop B** |
| --- | --- |
| Our normal price £475<br><br>Get 16% off our normal price | Get 15% off our normal price<br><br>Only pay £408 |

Which shop gives more money off their normal price?
Show your working clearly.

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 7*

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29 **8** Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

| Bank **G** | Bank **H** |
|---|---|
| 1.6% per year compound interest | 2.9% interest added after two years |

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?

..... Swiss francs

**(Total for Question 8 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 8*

30 **10** Johan owns a car.

The current value of the car is 32 000 euros.
The value of the car will now depreciate by 15% per year.

Lena is going to save up and buy Johan's car in 3 years' time.
She has already saved 2516 euros.

She will save an equal amount of money each month for the next 3 years.

Work out how much money Lena needs to save each month.

..... euros

**(Total for Question 10 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 10*

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**31** Each month Edna spends all her income on rent, on travel and on other living expenses.

She spends $\frac{1}{3}$ of her income on rent.

She spends $\frac{1}{5}$ of her income on travel.

She spends \$420 of her income on other living expenses.

Work out her income each month.

\$.....

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2018 · Question 13*

**32** **4** Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 $m^2$

Joshua buys one box of tiles for each 1.5 $m^2$ of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

.....%

**(Total for Question 4 is 5 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 4*

33

4 $N$ is a number.
17% of $N$ is 357

(a) Work out the value of $N$

$N =$ .....

(2)

n 2019, the population of a village was 650
n 2020, the population of the village was 806

) Work out the percentage increase in the population.

..... %

(3)

**(Total for Question 4 is 5 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 4*

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34

13 Antoine is going on holiday.
He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

| Payment | Amount paid |
|---|---|
| **Payment 1** | $\frac{3}{8}$ of the total cost |
| **Payment 2** | 45% of the total cost |
| **Payment 3** | \$406 |

Work out how much Antoine has to pay for Payment 2

\$ .....

**(Total for Question 13 is 5 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 13*

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35

**14** The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

**(Total for Question 14 is 5 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 14*

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**36** **5** Otis sells ice creams.

On Friday, Otis sells 75 ice creams.
On Saturday, Otis sells 87 ice creams.

(a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %

(3)

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

(b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs

(3)

**(Total for Question 5 is 6 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 5*

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**37** Teresa invests $2000 for 3 years in a savings account.
She gets 4% each year compound interest.

(a) How much money will Teresa have in her savings account at the end of 3 years?
Give your answer correct to the nearest dollar.

$.....

(3)

invested $T$
value of his investment decreased by 9% each year.

ıe end of the first year, the value of Sam's investment was $1365

Work out the value of $T$

.....

(3)

**(Total for Question 9 is 6 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 9*

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38 **6** In his previous job, Pierre was paid 400 euros in total for working a 5-day week.

In his new job, Pierre is paid 14 euros per hour.
In his new job, Pierre works for 7 hours each day for a 5-day week.

(a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week.

..... %

(4)

Marie changes her job.

Her salary decreases by 6%
Her new salary is 23 030 euros.

(b) Work out Marie's salary before she changes her job.

..... euros

(3)

**(Total for Question 6 is 7 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 6*

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## CHALLENGE QUESTIONS

**39**

**6** Shane bought a car.

The amount Shane paid for the car was $32 000

Theresa also bought a car.
To pay for this car, Theresa paid a deposit of $18 000 together with 14 monthly payments of $1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

.....%

(4)

Kylie bought a van.

After 1 year, the value of the van was $39 865
During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

$.....

(3)

**(Total for Question 6 is 7 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 6*

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**40** Osman buys a car for $16 000

The car depreciates at a rate of 12% each year for the first 2 years
In the third year, the car depreciates at a rate of $x$%

At the end of 3 years, the value of the car is $11 461.12

Work out the value of $x$

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 12*

**41** **13** Charlotte bought an apartment for $750 000

In the first year, the value of the apartment decreased by 4%
In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x$%

At the end of the third year, the value of Charlotte's apartment was $698 445

Work out the value of $x$

$x =$ .....

**(Total for Question 13 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 13*

**42** **13** Feruzi invests 80 000 Kenyan shillings (KES)
He invests the money for 3 years at $x$% compound interest each year.

At the end of 3 years, the total interest he receives is 6151.25 KES

Work out the value of $x$

$x =$ .....

**(Total for Question 13 is 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 13*

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**43** In September, the sales in a shop increased by 20%
In October, the sales in the same shop decreased by $x$%

Over these two months, the sales in the shop decreased by 10%

Work out the value of $x$

$x$ = .....

**(Total for question = 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 14*

**44** Simon bought a house at the beginning of 2018
The value of Simon's house had decreased by 15% by the end of 2018

The house increased in value during both 2019 and 2020
The percentage increases in the value of the house during 2019 and 2020 were the same.

The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018

Calculate the percentage increase in the value of the house during 2019

.....%

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 14*

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**TOTAL FOR PAPER IS 154 MARKS**